LASSONDE SCHOOL OF ENGINEERING | YORK U

ENG 1112 - Mini Projects: Coding and Linear Algebra

Project 1: Build and Break the Bracket

Student Guide

Thursday, October 29 to Friday, November 6, 2026 - Teams for the build and the test, individual for the sketch, the program and the reflection

1 What you are going to do

Over the next seven working days you will design a small bracket that could hold a solar panel off a wall, build it out of craft sticks, hang weight from it until it breaks, and then write a program that reads the test results from the whole class and turns them into a short engineering report.

That is three different kinds of work in one project. The first part is design: drawing something properly before you build it. The second is hands-on: cutting, gluing, loading, watching and writing down what happens. The third is coding: reading a data file and making sense of it with loops and if statements. The three parts fit together because the bracket you break on Tuesday is the one your program analyses on Wednesday, and the shape you draw this week is the kind of structure you will solve mathematically in Project 2.

Why this matters

Engineers rarely get to test a full-size structure to destruction. A cheap model that fails in a predictable way is how you learn where the weak points are before anything expensive is built. The sketch, the test record and the program are the three documents an engineer would produce from such a test, and this week you produce all three.

1.1 Teams

You work in a team of four (sometimes three or five), assigned at random within your lab. The teams are listed on the Project 1 teams page in the Mini Project 1 block on eClass. Find yours before the Thursday lab. You get a new team for each project, so this one lasts until Friday, November 6.

Done as a team	Done on your own
Agreeing on one design	Your own sketch of that design on the sketch template
Building the bracket	The program
Testing it on the rig and filling in the test record	The one-page output report from your program
Uploading the team's test data (one member uploads)	The half-page failure reflection
	The demo of your program to a TA

The team parts get one team mark. The sketch, the program, the output report and the reflection are marked for each student separately. Your program has to be your own: code copied from another student, including a teammate, is copying, and submitted code is checked for similarity.

Everyone in the team should have a stick in their hand during the build and a pen in their hand during the test. There is enough material for four people to work at once, and the test moves fast enough that one person keeping count of the bags, one lowering them in, one watching the joints and one writing is the right split.

1.2 The project at a glance

Day	What happens
Thursday, Oct 29	Course overview in the morning class, then Dr. van Breugel's first Python session. In the lab: meet your team, install Python, read the rules in Section 3, agree on a design and start your sketch on the template (Part A).
Friday, Oct 30	Dr. van Breugel presents Project 1 in full in class, then coding. The lab opens with a TA safety briefing (glue guns, cutting tools, the loading rig). A TA checks your team's design against the rules, then you build (Part B). Brackets stay in the lab until Tuesday.
Monday, Nov 2	Online. Finish your own copy of the sketch and read Part C so you know the test procedure before you get to the rig.
Tuesday, Nov 3	Morning lab: collect your bracket, add any string, glue and joint labels it still needs, and put it back in the box. Afternoon lab: load test on the rigs (Part C). Each team gets a 10-minute slot. One member uploads the team's CSV through the checker on eClass before leaving the lab, and in any case by 9:00 pm (Part D).
Wednesday, Nov 4	Online. Your personal data file and threshold appear on eClass in the morning. Write the program (Part E).
Thursday, Nov 5	Quiz 1 in the morning class. In the lab: finish the program and demo it to a TA. Your electronics kit for Projects 2 to 4 is handed out in class.
Friday, Nov 6	Hand-in: sketch, program, output report and reflection in the drop box by 9:00 pm.

2 What you get

Each team receives the materials in Table 1. Nothing else may go into the bracket, and unused material comes back at the end. The point is that every team builds from the same allotment, so the results can be compared.

Table 1. Materials per team.

Item	Amount and details
Jumbo wooden craft sticks	36 sticks, each 150 mm long, 18 mm wide, about 2 mm thick. You may cut them to length. Offcuts count as used material, and you do not get more.
Hot glue	Three glue sticks, dispensed at the shared glue gun station.

Item	Amount and details
PVA (white) glue	Available at the station as an alternative. Stronger joints, but it needs to dry overnight, which the Friday to Tuesday gap allows.
Cotton string	1 m. Tie a short loop of it around the load joint for the pail's hook (the loop is not a member). The rest is optional: string can only be pulled tight, so it can stand in for a member that is stretched by the load but not for one that is squeezed.
Foam board	One piece, 100 mm by 150 mm. This stands in for the solar panel.
Grid paper	Two sheets of 5 mm grid, for the full-size layout you build on.
Wax paper	A sheet to build on, so the bracket does not glue itself to the drawing.
Labels	Small round stickers for naming the joints.

Shared at the lab: glue guns, scissors and snips, rulers, protractors, safety glasses, and the four test rigs described in Part C. You do not need WHMIS training for anything in this project. The safety briefing at the start of the Friday lab covers everything you will handle.

3 The rules for your bracket

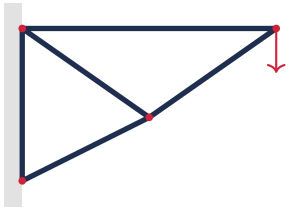
A bracket that breaks any of these rules will not be tested. A TA checks all nine on Friday before you build and again on test day.

1. **It must be a truss.** A truss is a structure made of straight pieces (called *members*) connected only at their ends (called *joints*, or *nodes* on the sketch template). Think of a railway bridge or a roof frame. Every piece of your bracket is a stick running from one joint to another.
2. **It must be flat.** All the members lie in one vertical plane that stands straight out from the wall, the way a shelf bracket does. Every stick lies flat in that plane with its wide face showing, and sticks overlap face to face where they meet. You may glue two sticks face to face to make one thicker member, and that pair counts as one member.
3. **It has a mounting face.** One straight vertical member, exactly 120 mm long, stands against the wall (the post on the test rig) along its long edge. Because the bracket is flat and sticks straight out from the wall, only that thin edge touches the post, and it is the only part the clamp holds. Figure 3 shows how.
4. **It reaches 200 mm.** The point where the load hangs is 200 mm out from the wall, measured horizontally. You have 10 mm either way.
5. **It has 5, 7 or 9 members**, counting the mounting face as one of them. Section 4.2 explains why 6 or 8 cannot work.
6. **Members only meet at their ends.** Glue goes on joints, not along the length of a stick. No member crosses another one except at a joint.
7. **The load hangs from a joint**, not from the middle of a stick. Put a label on that joint.

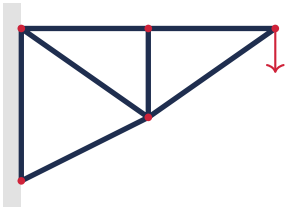
8. **The panel is not structural.** The foam board is glued to the top of the bracket at a tilt of 30 to 45 degrees from horizontal, as a solar panel would be. It carries no load in the test.

9. **Only the materials in Table 1.** No tape, no pins, no wire, no extra sticks.

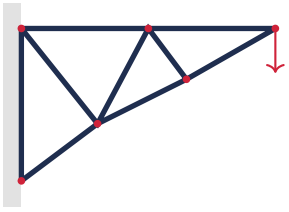
There is no limit on how much of the allotment you use, but the number of sticks in the finished bracket goes into the class data file as its mass, at 3 g per stick (Part B, step 9). A bracket that is strong and light is the better design.



Allowed. Five members, four joints. The smallest truss that passes the count.



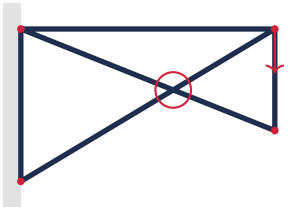
Allowed. Seven members, five joints. Triangles everywhere.



Allowed. Nine members, six joints. More joints to glue, shorter members.



Not allowed. Four members fail the count in Section 4.2, and a four-sided shape with no diagonal folds like a parallelogram anyway.



Not allowed. Two members cross without a joint where they meet.

Figure 1. Five designs against the rules. The top row shows the three member counts the rules allow, drawn without dimensions: where you put the joints is up to your team. The grey strip is the wall, the red arrow the hanging load. A triangle cannot change shape without a member changing length, which is why trusses are built from triangles.

Design advice

Trusses are strong because they are made of triangles. Before you commit to a design, trace it with your finger and ask at every four-sided shape: what stops this from folding? If the answer is "the glue", add a diagonal. Also think about which members get pushed and which get pulled when the load hangs. A member is pushed when the load tries to squeeze it shorter (engineers say it is in *compression*) and pulled when the load tries to stretch it (*tension*). A thin stick that is pushed along its length buckles sideways long before it crushes, so the pushed members are the ones to double up. You will calculate these forces in Project 2.

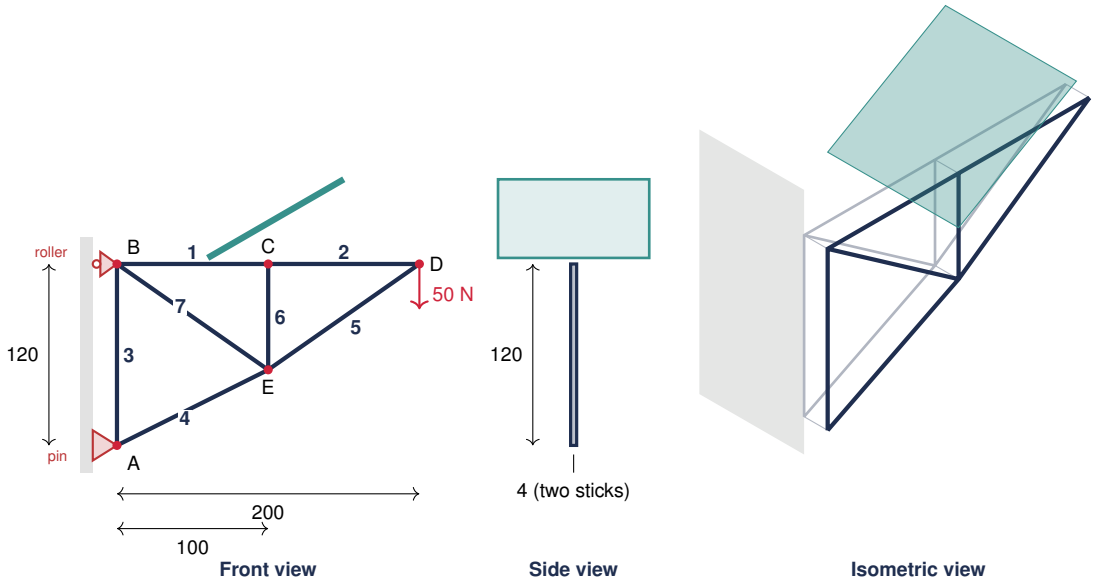


Figure 2. The three views of a bracket, as they go on sheet 1 of the template. Joints have letters, members have numbers, the pin and the roller are drawn at the wall joints, and the load arrow carries its size. The grey plane in the isometric view is the wall. The side view is a thin strip because every stick lies flat in the bracket's plane. The front and side views need dimensions.

4 Part A: Sketch it before you build it

Nothing gets glued until a TA has checked your team's design against the nine rules. The sketch is also your record of what was built, and a sketch with missing dimensions is useless to anyone who did not see the bracket, so it has to be complete. Every member of the team draws their own sketch of the agreed design on the two-page sketch template from eClass, and the sketch is handed in individually.

4.1 What a dimensioned sketch is

Engineers describe a three-dimensional object with several flat views. Figure 2 shows the idea. The *front view* looks straight at the flat face of the bracket, which is what you see standing beside it, and shows every member and joint. The *side view* looks at it end-on, from where the pail hangs back toward the wall, so a flat bracket appears as a thin strip about two sticks wide, which is exactly the point: that view proves the bracket is planar. The *isometric view* is the familiar three-dimensional picture, drawn with the three axes 120 degrees apart so that nothing is foreshortened.

4.2 Supports, load and the determinacy check

Sheet 2 of the template asks for three things that are new this week.

Supports. The wall holds the bracket at the two joints on the mounting face. On the rig, joint A sits on a ledge against the post, so it cannot move down or sideways: engineers call that a *pin*, and it provides two reactions (one horizontal, one vertical). Higher up, a clamp holds the face against the post, so joint B cannot move away from the wall but could slide along it: that is a *roller*, with one reaction. The number of reactions is therefore $r = 3$. Draw a small triangle under A for the pin and a triangle on a circle at B for the roller.

Load. Draw an arrow pointing down at the load joint and write the load beside it. Use 50 N, which is the 5 kg limit of the rig (5000 g is 49.05 N, and a sketch does not need the decimals).

Determinacy. Count your members m, your reactions r and your joints n, and check

$$m + r = 2n.$$

Each joint gives two equilibrium equations (horizontal and vertical), which is the $2n$. Each member force and each reaction is one unknown, which is the $m + r$. When the two sides match, the truss is *statically determinate*: every member force can be found from equilibrium alone, which is what you will do in Project 2. If $m + r$ is smaller than $2n$ the truss is a mechanism and will fold. If it is larger there is a redundant member, and the forces cannot be found without knowing how stiff each stick is. With $r = 3$, the check says: 4 joints need 5 members, 5 joints need 7, 6 joints need 9. All three fit the rules. The count is necessary but not the whole story: a four-sided panel plus a spare member passes the count and still folds, so the members also have to form triangles, which is what the TA looks for on Friday.

4.3 Steps

1. **Agree the design as a team** on scrap paper first. Decide the joint positions and which members are doubled.
2. **Draw the front view on sheet 1** of the template at half size: one small grid square is 10 mm of bracket, so the 200 mm reach is 20 squares and the 120 mm face is 12 squares. Draw each member as a single line and each joint as a small circle. Draw the pin, the roller and the load arrow.
3. **Label the joints** with capital letters. A is at the bottom of the mounting face and B at the top of it, so B is always at $(0, 120)$. Continue through the alphabet in any order. **Number the members** 1, 2, 3 and so on, in any order, and write the number beside each one.
4. **Draw the side view and the isometric view** in their boxes. The side view is at the same scale as the front view. The isometric view can be freehand, but it must show the panel.
5. **Add dimensions**: the reach, the height of the mounting face, the length of every member and the tilt angle of the panel. A dimension is a thin line with arrows at both ends and the number in the middle, as in Figure 2.
6. **Fill in sheet 2.** The nodes table takes each joint's x and y in millimetres, with the origin at A, x positive away from the wall and y positive upward, and a note of any support or load at that joint. The members table takes each member's number, its two end joints, its length and the stock piece used (one stick, or two doubled). Fill in the supports and load box and the determinacy check.
7. **Complete the title block**: team number, section and lab, date, drawn by (you), checked by (a teammate who has compared your sheet with theirs) and the scale (1:2). List the team members.
8. **Get the design checked.** On Friday, one team member brings a completed sketch to a TA before any glue is used. The TA checks the nine rules and the determinacy count.

Check yourself

Can someone who has never seen your bracket build an exact copy from your sketch alone? If a dimension is missing, they cannot. If the joint letters on the drawing do not match the nodes table, they will build the wrong thing. Fix both before the TA check.

5 Part B: Build it

1. Draw the front view again at full size on the grid paper: a 100 mm member is 100 mm long, which is 20 squares. This is your layout, not a deliverable, so it only needs the joints and the member lines.
2. Lay the wax paper over the full-size layout. You will build directly on top of it, so the joints end up exactly where you drew them.
3. Every stick lies flat on the paper, wide face down. At each joint the end of one stick lies on the end of the other, face to face, with about 15 mm of overlap, and that overlap is the glue joint. Nothing stands on its edge. Figure 3 shows the result from three directions.
4. Measure and cut every stick to length before gluing anything: the joint-to-joint length plus the overlap at each end. Check each cut piece against the layout. A member that is 5 mm short pulls the joint out of position and changes the geometry. Keep the outer edge of the mounting face straight and clean, because that edge is what sits against the post.
5. Glue one joint at a time. Put a small blob on the end of the stick, press it onto the joint, hold it for 20 seconds. For hot glue, aim for a small fillet, a little curve of glue, on both faces of the joint. A large blob adds mass and does not add strength.
6. If a member is doubled, glue the two sticks face to face first, then treat the pair as one stick.
7. Attach the foam panel last. It goes on the top member at a tilt between 30 and 45 degrees. Then tie a loop of string around the load joint, tight against the sticks, for the pail's hook.
8. Put the joint labels on, write your team number on the mounting face, and leave the bracket on the shelf the TA points you to. It stays in the lab until Tuesday, so hot glue and PVA both have time to reach full strength.
9. On test day, before your bracket is clamped, count the sticks in it: one for each single member and two for each doubled member, which is what the members table on sheet 2 of your sketch lists. There is no scale in the lab, so every stick counts as 3 g whether or not it was cut, and an eight-stick bracket is recorded as 24.0 g. Write the count and the mass on your test record. Take a photo of the bracket beside a sketch.

Careful

Hot glue guns run at about 190 degrees Celsius. Hold sticks by the far end, keep the gun on its stand between joints, and if you do get glue on your skin, go straight to the cold tap at the lab sink and tell a TA. Snips and craft knives stay on the bench, and you cut away from your hand.

6 Part C: Test it to failure

Test day is Tuesday. Read this part before you get to the lab: each team has a 10-minute slot at a rig, and the procedure has to be followed the same way by every team, because the whole class analyses the data you record.

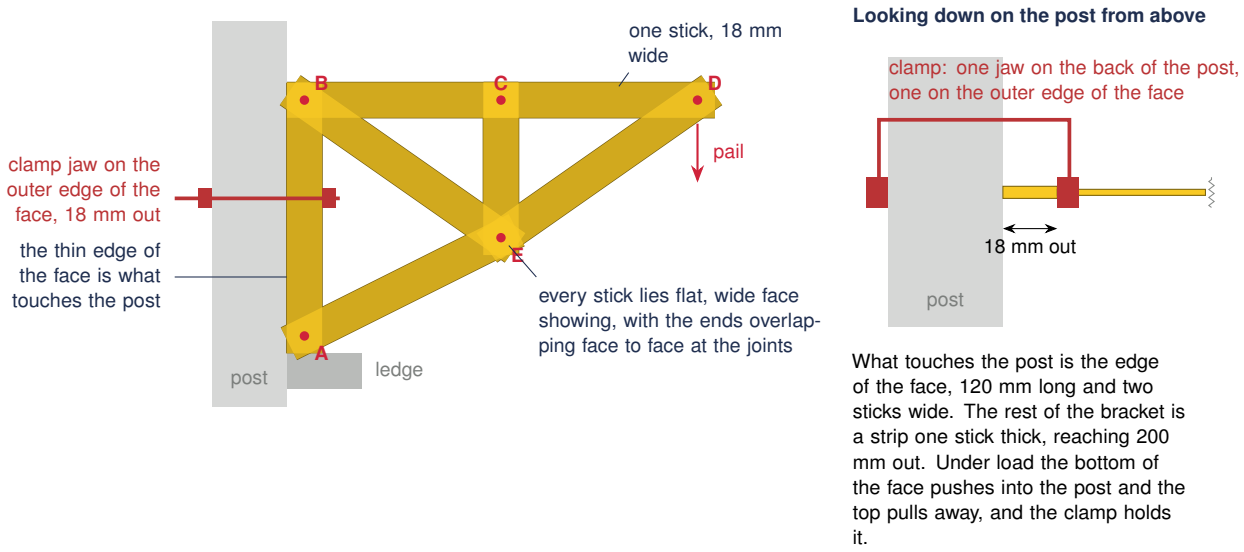


Figure 3. How the bracket meets the post. Left: from beside the rig, looking along the post, which is the front view on your sketch. Every stick shows its wide face. Right: from above, the bracket is a thin strip and the clamp squeezes the mounting face edge to edge.

6.1 The test rig

There are four rigs. Each one is a wooden post standing on a long foot that is clamped to the bench. Your bracket sits on a small ledge on the post, 300 mm above the foot, with the edge of its mounting face against the post. One screw clamp, about halfway up the face, squeezes the face against the post, one jaw on the back of the post and the other on the outer edge of the face. Under load the bottom of the face presses into the post above the ledge and the clamp holds the top. A pail hangs from the load joint, 200 mm out from the post, a few centimetres above the foot, so that it has only a short drop when the bracket fails. You add sand to the pail one bag at a time until the bracket fails.

Each rig has ten bags of sand, and every bag holds 500 g. There is no scale. The load on the bracket is the number of bags in the pail times 500 g, and that is what you record.

6.2 Loading

Everyone at the rig puts on safety glasses before the pail is hung. Sticks snap with some energy and splinters travel. Once the pail is hanging, hands stay clear of the bracket and the pail except to lower a bag in. The test runs until the bracket fails or the tenth bag is in, and loading stops at once if the TA says so.

1. Clamp the bracket. The edge of the mounting face goes against the post with joint A resting on the ledge. A TA tightens the screw clamp about halfway up the face, so that joint A stays on the ledge and the face sits flat against the post. Check that nothing else on the bracket touches the post.

2. Hang the empty pail from the load joint with the S-hook. Take a photo of the loaded rig from the front. The pail and hook are the same for every team and are not counted: the load you record is the sand alone.

3. This is **step 0**. Its row in the test record (Appendix B) is already filled in: cumulative mass 0.0, status `HELD`, observation "pail only".

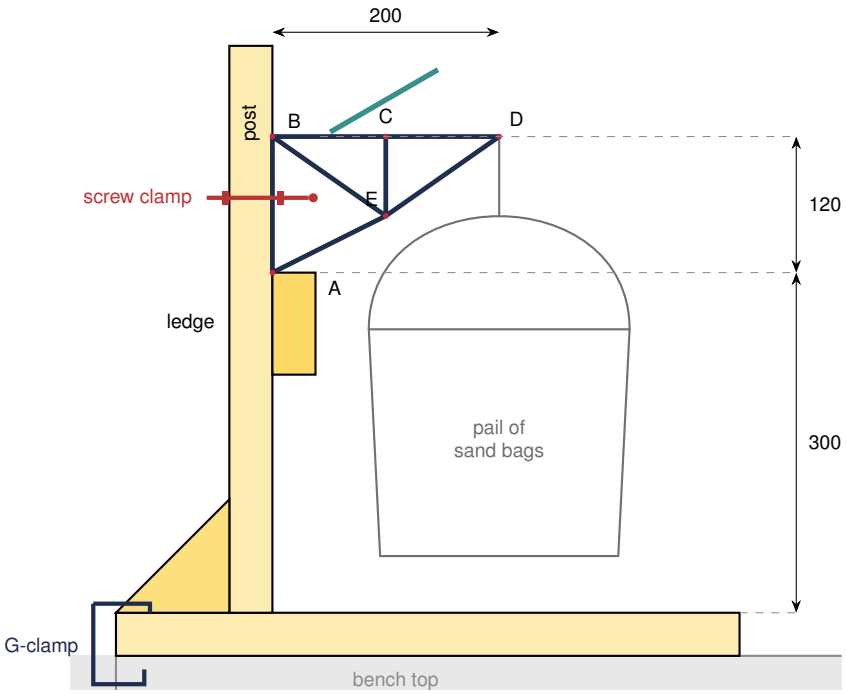


Figure 4. The test rig from the front. Joint A sits on the ledge, 300 mm above the foot, so every team's bracket is in the same place. The pail hangs from the load joint a few centimetres above the foot, so it has only a short drop when the bracket fails.

4. Now repeat the following for each step:
   - Move to the next row. The *cumulative mass* column is already filled in, because every bag is 500 g: it reads 500.0 at step 1, 1000.0 at step 2, and so on, 500 times the step number. (Cumulative means "everything so far".) Check that the number of bags about to be in the pail matches the step number.
   - Lower one bag gently into the pail. Do not drop it: a dropped bag applies a shock load that is much larger than its weight.
   - Wait 10 seconds. Watch the joints and the longest members that are being pushed.
   - Write one observation. "Nothing yet" is a valid observation, and so is "member 5 (ED) starting to bow" or "joint E creaking".
   - Write status `HELD` and go to the next step.
5. **Stop and write `FAILED`** the moment any one of these happens: a member breaks, a member buckles visibly, a joint separates, the panel comes off, the mounting face slips in the clamp, or the load joint drops more than 20 mm from where it started. In the *mode* column write the code from Table 2, and in the observation column name the member (number and end joints) or the joint.
6. **Stop and write `CAPPED`** if the tenth bag is in and the bracket is still holding. That is 5000 g, the limit of the rig, and you have reached it. Write mode `NF`.
7. Take a photo of the failure before touching anything. Then the TA removes the pail, then the clamp, then the bracket.

8. Read through the test record as a team before leaving the rig. Every row up to the last one needs a status and an observation, and the last one needs a mode code as well. A TA initials the sheet. Put all ten bags back beside the rig for the next team.

Table 2. Failure mode codes. Pick the one that happened first.

Code	Name	What it looks like
`JS`	Joint separation	A glued joint opened. Both sticks are still intact. The glue let go.
`MB`	Member buckling	A stick that was being pushed along its length bowed sideways and either snapped or folded. The classic truss failure.
`MF`	Member fracture	A stick broke, usually at a knot or where it was cut, while being pulled or bent.
`MS`	Mounting slip	The mounting face slid in the clamp. Not really a bracket failure, but it ends the test.
`PD`	Panel detachment	The foam board came off before anything structural failed.
`DL`	Deflection limit	The load joint sagged past 20 mm but nothing visibly broke. The bracket is too flexible to be useful.
`NF`	No failure	Only with status `CAPPED`.

If two things seem to happen at once, record the one you think happened first and say so in the observation.

6.3 Converting to newtons

Mass in grams is what you record. Force in newtons is what the bracket feels. The two are related by gravity:

$$F\ (\text{newtons}) = \frac{m\ (\text{grams})}{1000} \times 9.81.$$

One bag therefore pulls down with $0.5 \times 9.81 = 4.9$ N, and six bags, 3000 g, with $3.0 \times 9.81 = 29.4$ N. Your program does this conversion. Your **maximum sustained load** is the cumulative mass at the last `HELD` row, converted to newtons. The `FAILED` row's mass does not count: that load was applied but the bracket did not hold it. The true failure load lies somewhere between the two, so it is known to within one bag, about 5 N. That is the resolution of this test, and it is the same for every team.

7 Part D: Record and upload the data

7.1 What a CSV file is

A CSV (comma-separated values) file is a plain text file where each line is one record and the values in the line are separated by commas. It is the simplest possible spreadsheet, and any program can read it. Open one in Notepad or TextEdit and you see the raw text. Open it in Excel and you see a table. Your program will read the raw text.

Your team's file has one line per load step and looks like this:

```
team,step,cum_mass_g,status,mode,members,bracket_mass_g,note
07,0,0.0,HELD,,7,24.0,pail only
07,1,500.0,HELD,,7,24.0,nothing yet
07,2,1000.0,HELD,,7,24.0,nothing yet
07,3,1500.0,HELD,,7,24.0,slight bow in member 5 (ED)
07,4,2000.0,FAILED,MB,7,24.0,member 5 (ED) buckled at midspan
```

The first line is the *header*: it names the columns and is the same for every team. Each line after that is one step.

Column	What goes in it
`team`	Your two-digit team number, for example `07`.
`step`	0, 1, 2, and so on.
`cum_mass_g`	Cumulative mass of sand in grams, to one decimal place: 0.0 on step 0, then 500 times the step number. The pail is not counted.
`status`	`HELD`, `FAILED` or `CAPPED`. Your last row is `FAILED` or `CAPPED`. Every other row is `HELD`.
`mode`	The failure code, on the last row only. Leave it empty on `HELD` rows (two commas in a row, as in the example).
`members`	Your member count. The same number on every row.
`bracket_mass_g`	The bracket's mass from Part B, 3 g per stick, to one decimal place. The same number on every row.
`note`	Your observation. **No commas inside the note**, because a comma would start a new column.

7.2 Upload

The file lives inside the activity called `team_template` in the Mini Project 1 block. Open the activity and you find `team_template.csv` with the example rows above already in it. Replace every line except the first with your own rows, either in the editor on the page or by uploading a file of your own with the same name, `team_template.csv`. If you prepare it in Excel, choose "CSV (comma delimited)" when saving, not the default Excel format, and check that your team number still has its leading zero (Excel likes to turn `07` into `7`).

One member of the team submits for everyone. It is due by 9:00 pm on Tuesday, November 3, and the best time to do it is at the rig, before you leave the lab, while the whole team can still check the numbers.

Press **Evaluate** and eClass checks the file on the spot and tells you what, if anything, is wrong: a header that does not match, a missing column, a team number without its two digits, a step out of sequence, a mass that did not go up, a status that is misspelled or in lower case, a `FAILED` row with no mode code, an empty note. Fix the file and evaluate again. You can do that as many times as you like before 9:00 pm, and only the last submission counts. The check does not look for a comma inside a note, so look for that yourselves.

The teaching team merges every accepted file into one class file that evening. A file that was

never accepted is not in it, which means your own team's data is missing from the file you analyse.

8 Part E: Write the program

This part is individual. The language is Python 3, the same as ENG 1111.

8.1 What the program does

On Wednesday morning you download two things from the Mini Project 1 block: a **personal data file** and a **personal safety threshold** in newtons. The data file is the class file with some teams left out and the order shuffled, and the threshold is a number, both chosen from your student number. No two students get the same file and threshold, so your correct output is different from your neighbour's.

Your program reads the data file and prints a report with four parts:

1. **Maximum sustained load per team**, in newtons, one line per team, in the order the teams appear in your file.
2. **The class average** of those maxima. You must compute it with an *accumulation loop*: a running total that you add to, a counter that you increment, and one division at the end.
3. **A count of failures by mode code**, one line per code that appears at least once. A `CAPPED` bracket counts as `NF`.
4. **A flag for every team** whose maximum sustained load is below your safety threshold, with the shortfall in newtons.

8.2 The rules

- Use variables, conditionals (`if`, `elif`, `else`) and loops (`for`, `while`) only, together with the file-reading and printing statements from the coding notes on eClass. Built-in functions such as `max()`, `sum()` and `sort()` are not permitted for the four computations above. The point of the project is to write the loop yourself.
- Nothing from the data may be typed into the program. No team numbers, no masses, no number of rows. The TA will run your program on a different file at the demo, and it has to work unchanged.
- The threshold is typed in by the user when the program starts (`input()`), not written into the code.
- Only `HELD` rows count toward the maximum.

8.3 How to think about it

The file is sorted by team, so all of team 07's rows come together, then all of team 23's, and so on. That means you can read the file once, top to bottom, and notice when the team number changes. Figure 5 shows the shape of the program.

The trickiest idea is in the box on the right: you only know a team is finished when you see the *next* team's first row. That is why the last team in the file needs to be finished separately, after the loop ends, because there is no next team to trigger it.

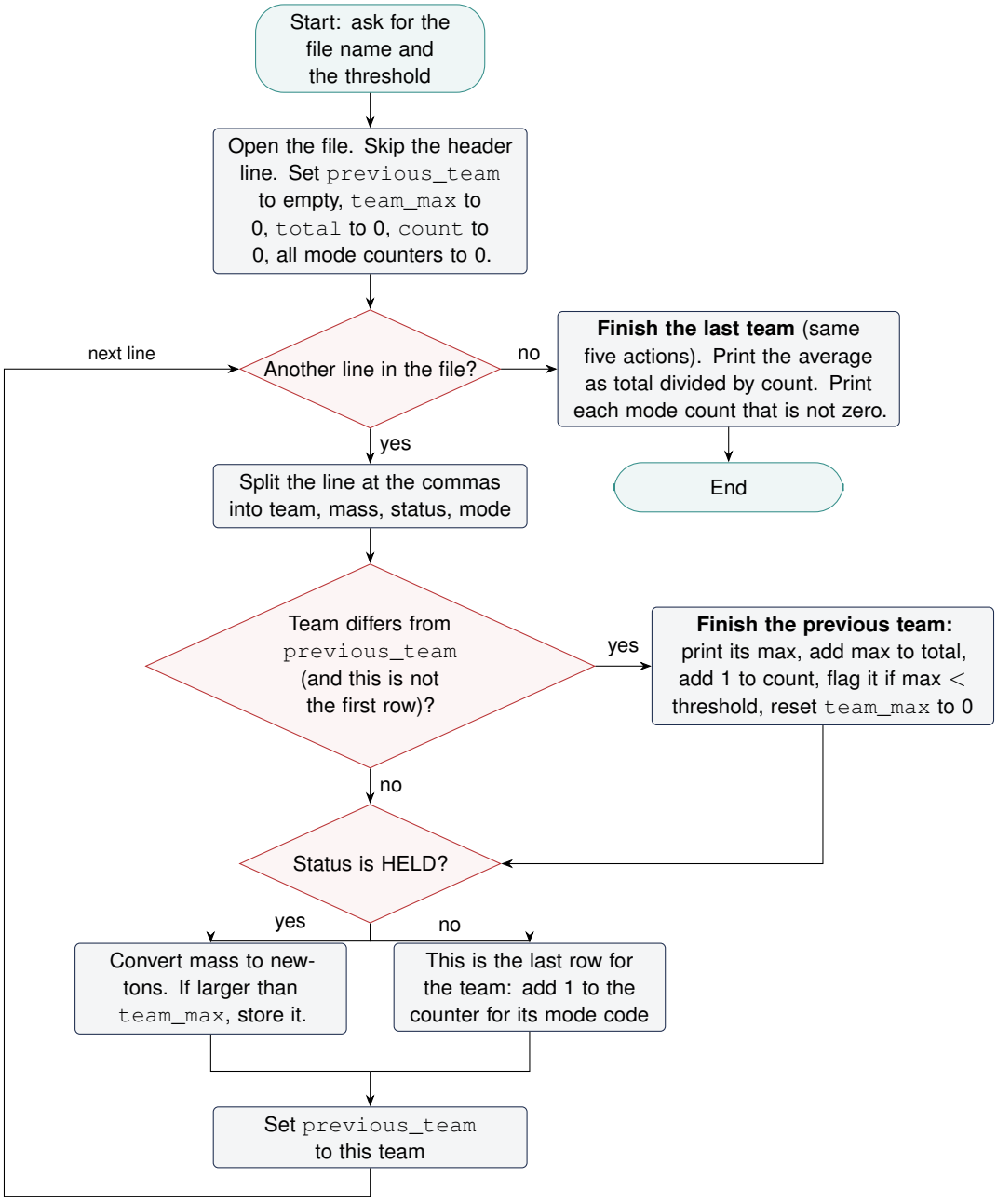


Figure 5. The shape of the program. One pass through the file, with a "team changed" check on every row.

8.4 A worked trace

Table 3 follows the program through a seven-row file with two teams and a threshold of 5.0 N. Follow it line by line and the variables will make sense.

Table 3. Trace of the loop on a small file. Threshold 5.0 N.

Row	team, mass, status, mode	previous	team_max	total	count	What happens
1	07, 0.0, HELD,	empty	0.00	0	0	First row, no team change. HELD, so 0.00 N stored.
2	07, 500.0, HELD,	07	4.91	0	0	HELD and larger, stored.
3	07, 1000.0, FAILED, MB	07	4.91	0	0	Not HELD. MB counter becomes 1. Max unchanged.
4	23, 0.0, HELD,	07	0.00	4.91	1	Team changed: print "07: 4.91 N", flag 07 (0.09 below 5.0), add to total, reset. Then HELD: 0.00 stored.
5	23, 500.0, HELD,	23	4.91	4.91	1	Stored.
6	23, 1000.0, HELD,	23	9.81	4.91	1	HELD and larger, stored.
7	23, 1500.0, FAILED, DL	23	9.81	4.91	1	DL counter becomes 1.
end				14.72	2	Loop over. Finish team 23: print "23: 9.81 N", no flag. Average $14.72/2 = 7.36$ N. Print MB 1, DL 1.

8.5 Getting started

Start from an empty file called `p1_bracket.py`. Ask for the file name and the threshold with `input()`, open the file, read the header line and ignore it, then loop over the remaining lines, splitting each one at the commas. Build the program in this order and run it after each step:

1. Print each team's maximum. Check against the file by hand for one team.
2. Add the total and count, and print the average.
3. Add the mode counters.
4. Add the threshold flag.
5. Run it on a second file (any classmate's works) to make sure nothing is hard-coded.

Common mistakes

Forgetting to finish the last team after the loop. Using the `FAILED` row's mass as the maximum. Comparing a string to a number (everything read from a file is text until you convert it with `float()`). Dividing by 9.81 instead of multiplying. Forgetting that a `CAPPED` row is not a `HELD` row: its mass is not a sustained load, and its mode `NF` is counted like any other code.

8.6 The demo

In the Thursday lab a TA gives you a fresh data file and a threshold. You run your program on it, which takes about thirty seconds. Then the TA points at a line of your code and asks what it does and why it is there, and does the same for every member of your team. You should be able to explain any line, because you wrote all of them.

9 What to hand in

Your four files go in the drop box called **Project 1 submission (individual)** in the Mini Project 1 block by 9:00 pm on Friday, November 6. The team's CSV goes through the `team_template` checker in the same block by 9:00 pm on Tuesday, November 3, uploaded by one member for the whole team.

Item	What it is
Sketch (you)	Both sheets of the template, complete and with the title block filled in, scanned or photographed into one PDF.
Program (you)	`p1_bracket.py`, the Python file itself.
Output report (you)	One page: what your program printed for your personal file and the threshold you used, with your name and student number.
Reflection (you)	Half a page, with your failure photo. What broke, why it broke there and not somewhere else, which members were in tension and which in compression, and one change worth trying. Name members by number and end joints.
Test data (team)	`team_template.csv`, filled in, through the `team_template` activity, Tuesday by 9:00 pm. The paper test record with the TA's initials stays with the team.
Demo (you)	In the Thursday lab, with a TA.

10 How it is marked

Project 1 is worth 15 percent of the course grade, with the reflection marked as part of it. Each specification below is either met or not yet met, with no part marks. Late work loses 15 percent per calendar day for up to three days. Quiz 1 is separate, written on paper in the Thursday morning class, and covers the ideas in this project.

Level	Specifications
Core	Sketch complete on both sheets, with the determinacy check done. Test record complete with all rows filled and the CSV accepted by the checker. Program prints all four parts correctly for your personal file and runs unchanged on the demo file. Reflection names a specific member or joint.
Higher	Program handles a `CAPPED` team correctly. Flag lines include the shortfall. Reflection explains the failure in terms of which members were pushed and which were pulled, and proposes a change that addresses it.

11 Getting help

TAs are in the lab for every session. Ask early. Teams run out of time for two reasons: a design that was not triangulated and had to be rebuilt on Friday afternoon, and a bug in the "team changed" logic that fifteen minutes with a TA on Thursday would have found.

11.1 Words used in this guide

Word	What it means here
Truss	A structure made of straight members joined only at their ends.
Tension, compression	A member being pulled is in tension, a member being pushed is in compression. Tension is positive in this course.

Appendix A: Checklist before you leave the rig

☐ Every row up to the last one has a status and an observation.

☐ Exactly one row says `FAILED` or `CAPPED`, and it is the last one.

☐ That row has a mode code and names a member or joint.

☐ Three photos taken: unloaded, loaded, failure. The failure photo goes in your reflection.

☐ Stick count and bracket mass written at the top.

☐ The CSV is uploaded and the checker said OK.

☐ All ten bags back beside the rig for the next team.

Appendix B: Test record

Team _____ Rig _____ Members _____ Sticks _____ Bracket mass (g) _____ TA initials _____

Bracket mass is 3 g per stick. Every bag is 500 g, so the cumulative mass is already filled in: check it as you go. The pail is not counted. Status is `HELD` on every row except the last, which is `FAILED` with a mode code or `CAPPED` with `NF` after the tenth bag. Rows after the failure stay empty.

Step	Cum. mass (g)	Status	Mode	Observation
0	0.0	HELD		pail only
1	500.0			
2	1000.0			
3	1500.0			
4	2000.0			
5	2500.0			
6	3000.0			
7	3500.0			
8	4000.0			
9	4500.0			
10	5000.0			

Mode codes: `JS` joint separation, `MB` member buckling, `MF` member fracture, `MS` mounting slip, `PD` panel detachment, `DL` deflection limit (load joint down more than 20 mm), `NF` no failure (with `CAPPED` only). In the observation on the last row, name the member (number and end joints) or the joint.

Photos: ☐ unloaded ☐ loaded ☐ failure CSV uploaded and accepted: ☐